Project Title

Milestone: Project proposal, Conceptual modeling, etc.

Group 7
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Percentage of Effort Contributed by Student1: 50%

Percentage of Effort Contributed by Student2: 50%

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Title of the Project: Restaurant Management

Business Problem Statement:
Restaurants face many operational challenges in managing things like customer flow, staffing allocation, and service quality. Most systems lead to inaccurate wait time estimates, unbalanced server workloads, lost reservations, and poor table utilization, which results in customer dissatisfaction, revenue loss, and staff frustration.

The Problem:
Restaurants need an integrated data management system that coordinates waitlist tracking, reservation management, table assignments, and server workload balancing while capturing guest preferences and enabling performance analytics.

Other Requirements:

1. GUEST and RESERVATION: A guest can make zero to infinite reservations(for different dates/times); a reservation must be made by exactly one guest.
2. GUEST and GUEST_NOTE: A guest can have zero to infinite notes recorded by staff; a guest note must belong to exactly one guest
3. TABLE and SECTION: A table must belong to exactly one section; a section can contain multiple but must have at least one in its section.
4. SERVER and SECTION: A server must be assigned to exactly one section per shift; a section can be assigned to multiple but at least one server across different shifts.
5. SEVER and SHIFT: A server can work zero to infinite shifts; a shift must be worked by at least one server.
6. SEATING_ASSIGNMENT and TABLE: A seating assignment must be exactly one table; a table can have zero to infinite seating assignments over time(different parties throughout service).
7. SEATING_ASSIGNMENT and SERVICE: A seating assignment must be served by exactly one server; a server can have zero to many seating assignments during their shift.
8. SEATING_ASSIGNMENT and RESERVATION: A reservation can result in zero or one seating assignment; a seating assignment can be linked to zero or one reservation.
9. GUEST and SEATING_ASSIGNMENT: A guest can have zero to infinite seating assignments; a seating assignment must be before exactly one guest(or party).
10. SERVER and PERFORMANCE_METRICS: A server can have zero to infinite performance metric records; a performance metric record must belong to exactly one server.
11. TABLE and STATUS: A table must have exactly one current status at any time; a status type (available, occupied, reserved, bussed) can apply to zero to infinite table simultaneously.